Customer Shopping Behavior Analysis

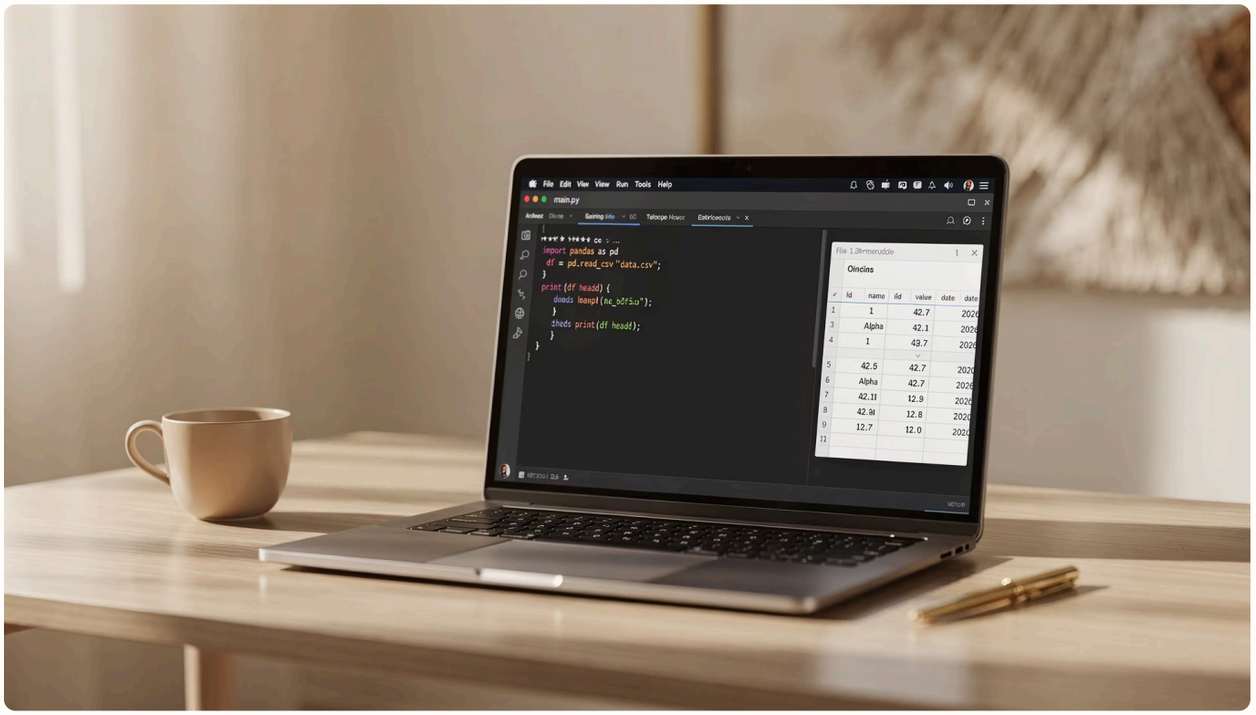
Uncovering insights from **3,900 transactions** across product categories, demographics, and spending patterns.

Dataset at a Glance

3,900	18	$59.76	3.75
Purchases	Columns	Avg Spend	Avg Rating
Transactional records analyzed	Features per record	Average purchase amount	Average review rating

Key features: demographics, purchase details, shipping, discounts, and subscription status.

Data Preparation in Python



01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Standardize

Imputed missing `Review Rating` with category median; renamed to `snake_case`

03

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

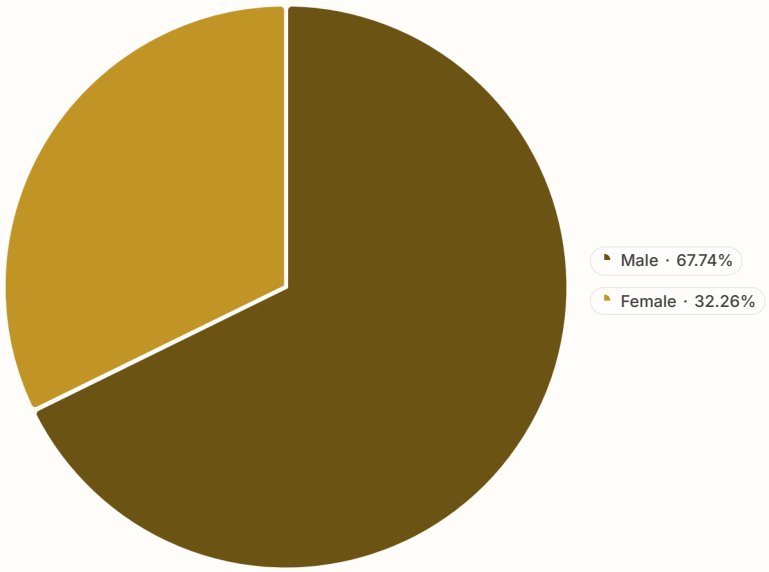
04

DB Integration

Loaded cleaned data into **PostgreSQL** for SQL analysis

SQL ANALYSIS

Revenue by Gender & Discount Users



High-Spending Discount Users

839 customers used discounts yet spent **above average** — a key segment for targeted retention.

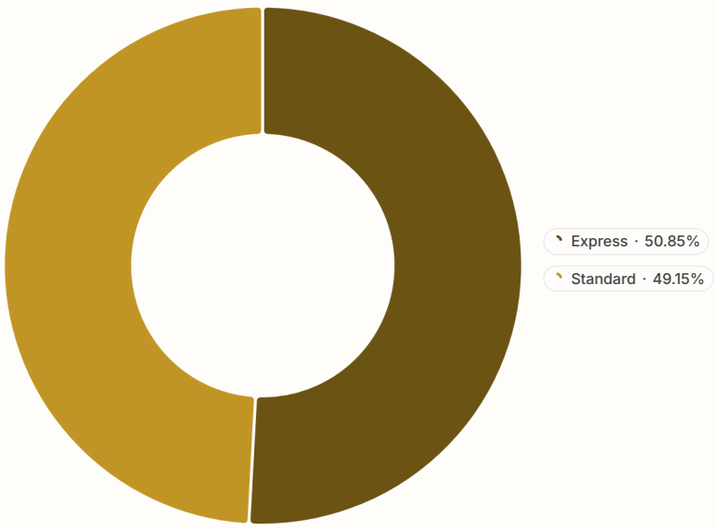
Discount users are not low-value — many are high spenders worth nurturing.

SQL ANALYSIS

Top 5 Products by Rating

1	Gloves ⭐ 3.86
2	Sandals ⭐ 3.84
3	Boots ⭐ 3.82
4	Hat ⭐ 3.80
5	Skirt ⭐ 3.78

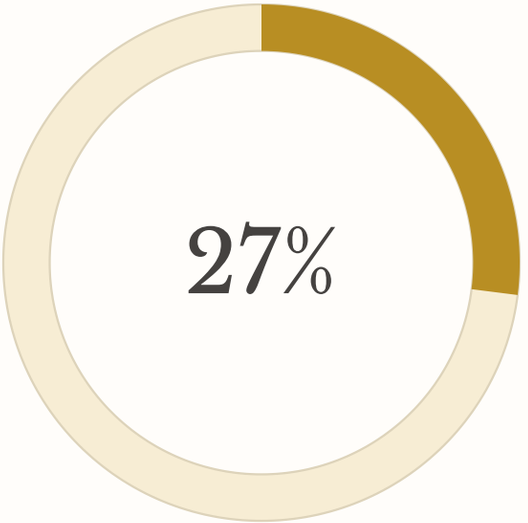
Shipping: Standard vs. Express



Express shipping users spend slightly more on average.

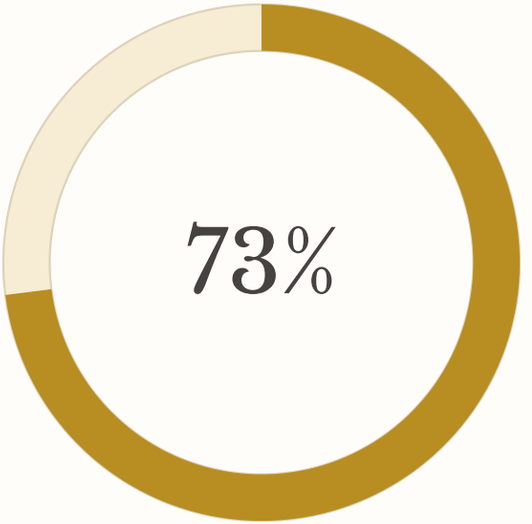
SQL ANALYSIS

Subscribers vs. Non-Subscribers



Subscribers

1,053 customers · $62,645 revenue



Non-Subscribers

2,847 customers · $170,436 revenue

Avg Spend is Nearly Equal

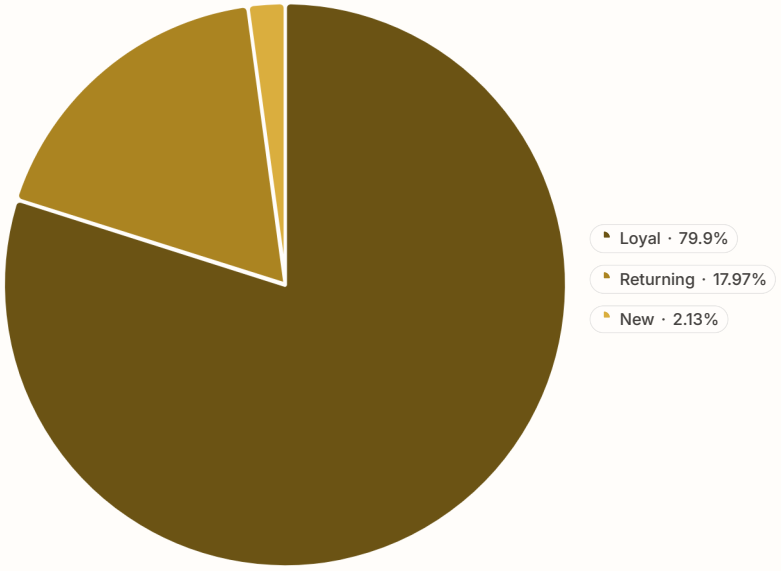
Subscribers: **$59.49** | Non-subscribers: **$59.87**

Despite similar spend per transaction, non-subscribers dominate total revenue due to sheer volume.

Opportunity: convert non-subscribers to unlock loyalty benefits.

SQL ANALYSIS

Customer Segmentation & Loyalty



Repeat Buyers & Subscriptions

Of customers with **>5 purchases**:

Most loyal buyers are **not yet subscribers** — a clear conversion opportunity.

2,518

Not subscribed

958

Subscribed

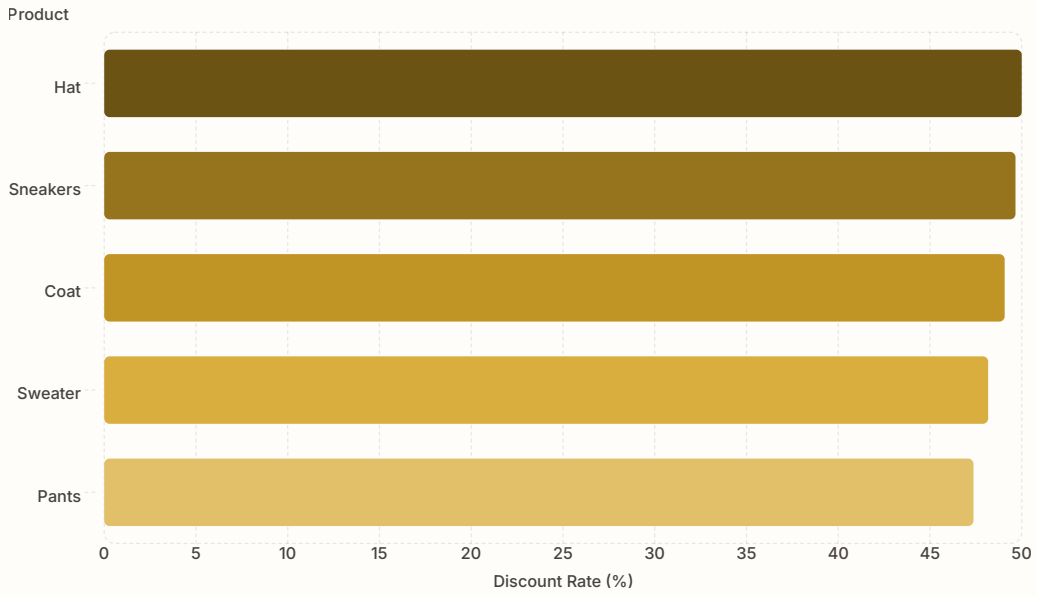
SQL ANALYSIS

Top Products & Discount Dependency

Top 3 per Category

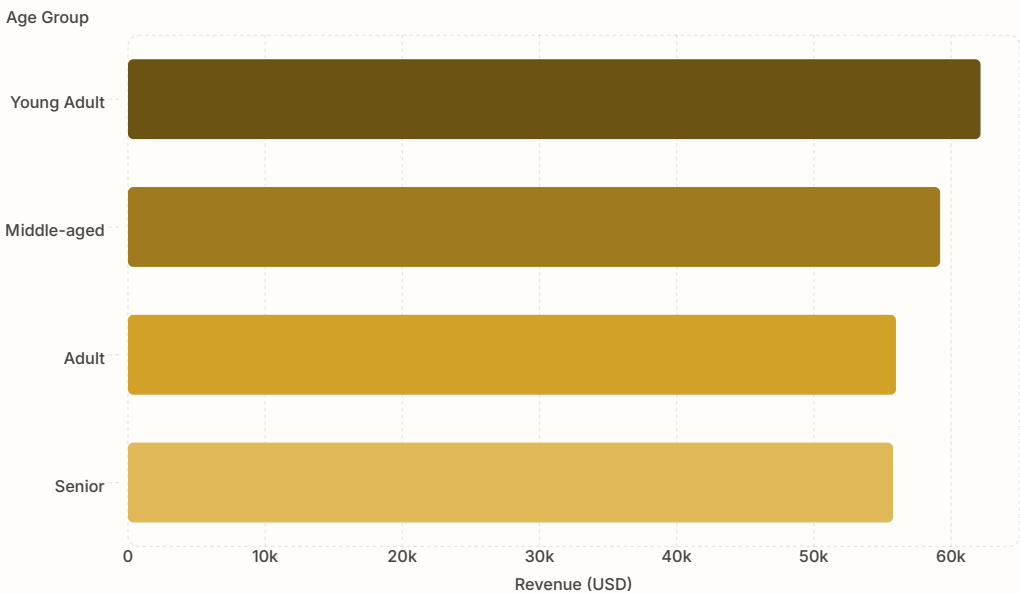
Category	Top Items	Orders
Accessories	Jewelry, Sunglasses, Belt	171, 161, 161
Clothing	Blouse, Pants, Shirt	171, 171, 169
Footwear	Sandals, Shoes, Sneakers	160, 150, 145
Outerwear	Jacket, Coat	163, 161

Most Discount-Dependent Products



SQL ANALYSIS

Revenue by Age Group



Young Adults Lead Revenue

Young Adults contribute the most at $62,143, followed closely by Middle-aged customers.

Revenue is relatively balanced across all age groups — no single group dominates.

Power BI Dashboard

Interactive dashboard consolidating subscription status, revenue by category, sales by age group, and key KPIs in a single view.

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → move to "Loyal"

Review Discounts

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Product Positioning

Highlight top-rated & best-selling items in campaigns